

Maryam Amir

maryamamir7865@gmail.com | +92 319 8050754 | <https://github.com/MA7865>

OBJECTIVE

- Second-semester Computer Science student at NUST SEECS (CGPA: 3.51).
- Research-internship experience in agentic AI and data science at TUKL-DL Lab — demand forecasting, computer-vision-assisted product matching, real-time data streaming, and bot detection systems.
- Independently built and shipped applied-AI and full-stack projects outside research — an academic scheduling system, and an offline RAG-based chatbot — reflecting a growing focus on machine learning, computer vision, and backend/agentic systems.
- Eager to contribute to interdisciplinary projects that encourage curiosity, teamwork, and innovation.

ACADEMIC QUALIFICATIONS

Bachelors in Computer Science — NUST, Islamabad 2025 – 2029

- In progress (CGPA: 3.51)

Bachelors in Electrical Engineering — NUST, Islamabad 2024 – 2025

- Completed first year with a CGPA of 3.84 / 4.00 before transitioning to Computer Science.
- Worked on engineering projects including a solar panel cleaning system and a smart waste management solution, gaining hands-on experience with hardware design and control systems.

A-Levels — Super Nova College, Islamabad 2022 – 2024

- Mathematics, Physics, Computer Science, Chemistry — 4 A*s

IGCSE — Ajyal Al Mareefa International School, Jizan, Saudi Arabia 2020 – 2022

- Maths, Physics, Chemistry, Biology, Computer Science, English, Urdu, Islamiyat, Pak.Studies – 8A*1A

EXPERIENCE

AI Training Analyst — Vue (Freelance/Part-time) August 2026 - Present

Remote

- Working on evaluating and improving the quality of AI-generated responses through systematic side-by-side assessments.
- Fact-checking model outputs, identifying hallucinations, and providing evidence-based evaluation reports to assess response quality.
- Applying evolving quality-assurance guidelines and evaluation criteria to improve model performance and consistency.

Research Intern — HPC Lab, SEECS, NUST July – August 2026

8-weeks paid internship (Large Language Models & Efficient Inference) | Supervisor: Dr. Sohail Iqbal | H-12, Islamabad

- Conducted an extensive literature review on **Large Language Model (LLM) inference acceleration**, focusing on speculative decoding and hierarchical speculative decoding techniques.
- Analysed and compared state-of-the-art decoding frameworks including **Speculative Decoding, Medusa, EAGLE, Cascade Speculative Drafting, SpecRouter, PipeSpec, CAS-Spec and HiSpec**.
- Identified architectural limitations and research gaps related to draft model quality, verification overhead and scalability in transformer-based inference.
- Reproduced baseline speculative decoding experiments using **Qwen2.5 models** and evaluated the effect of speculation length on inference speed.
- Investigated hierarchical speculative decoding as a potential research direction for lossless LLM acceleration.

Research Intern — TUKL-DL Lab, SEECS, NUST June – August 2025

10-weeks internship | H-12, Islamabad

- Conducted **data understanding, cleaning, and feature engineering** on five large Walmart/Amazon datasets — handling missing values, encoding categorical features, and identifying key demand-driving variables across sales trends, product attributes, and competitor pricing.
- Built the forecasting, product-matching, real-time streaming, and bot-detection components of a **Dynamic Pricing Agent** (see Projects) designed to forecast prices, predict demand, and guard against competitor price scraping.

- Built **demand forecasting** pipelines using GluonTS, Darts, Prophet, XGBoost, and ARIMA to model weekly sales per department across 45 Walmart stores, while controlling for exogenous covariates (holidays, promotions, competition effects).
- Developed a **cross-platform product-matching system** using sentence-transformers (all-mpnet-base-v2), NLP-based text similarity, image comparison, brand validation, and fuzzy matching to generate composite matching scores; when CAPTCHA blocks stopped automated scraping, manually curated an exact-match dataset to fine-tune the model instead.
- Configured Apache Kafka with 1,000 partitions (mapped on product ASINs) and used Apache Avro serialization to build a **real-time streaming pipeline for price and sales data**, setting up producers, consumers, brokers, and topics.
- Designed a **bot detection and sandboxing layer** classifying good vs. bad bots, deploying JavaScript challenges and cookie checks to prevent competitor price scraping.
- Explored **product-ranking analysis**: queried OpenAI with product prompts (e.g. "quiet ceiling fan"), scraped the top 5 returned links, extracted 15 ranked products, and compared rank position against attributes (ratings, reviews, price, SEO keywords) to find patterns in search visibility.
- Completed a Deep Neural Networks course and built a digit-classifier neural network. Attended seminars on quantum computing, AI governance, and AI ethics.
- Created and delivered four structured presentations — two evaluation presentations (mid/final) and two supervisor progress updates — covering Kafka architecture, demand forecasting models, and AI governance considerations.

Teacher Assistant (Maths & Physics) — Super Nova College, F-8

2022 – 2023

- Assisted students in problem-solving and concept clarification; conducted group lectures and shared academic feedback with faculty.

Community Teaching Volunteer — Hope Pakistan

September 2023

H-11 Slum School (behind FAST University)

- Taught underprivileged students for 30+ community service hours, received a Certificate of Appreciation

Participant — WWF Eco Internship Program

January 2022

- Completed 25 community service hours (highest in school) and presented a sustainability plan promoting eco-friendly waste management.

PROJECTS

DFT²W 2026 Conference Assistant Chatbot

Sept 2026

Team project (2 members) | Java, CSS, RAG, LLMs

- Developed a chatbot to improve the accessibility and visibility of IEEE Digital Futures and Transformative Technologies Week (DFT²W 2026) hosted at NUST SEECS.
- Built an AI assistant capable of answering visitor questions about the conference, including dates, research tracks, participation, workshops, and partnerships, using information from the official conference program.
- Collaborated with a fellow SEECS student to develop and integrate the chatbot into the conference's web presence.

Exploring Speculative Decoding Techniques for Accelerated LLM Inference

Aug – Sept 2026

Research Report | HPC Lab SEECS

- Conducted a literature review of Speculative Decoding, Medusa, EAGLE, Cascade Speculative Drafting, SpecRouter, PipeSpec, CAS-Spec, and HiSpec for lossless LLM inference acceleration.
- Analysed limitations related to draft-model quality, verification overhead, and scalability, narrowing the research direction toward Hierarchical Speculative Decoding.
- Investigated the trade-off between additional drafting and verification levels and their impact on end-to-end LLM inference latency.

SlotSync — Academic Appointment & Scheduling System

2025

Team project (4 members) | Java, JavaFX, MySQL

- Built to replace informal appointment booking over WhatsApp/email at NUST, which regularly caused professor double-bookings and wasted slots due to last-minute cancellations for both parties.
- Worked on the student view: the DB layer (OOP class schema, DAO queries) and a JavaFX front end for booking, viewing, and cancelling appointments.
- Designed a waitlist mechanism: when a student cancels last-minute, the freed slot is automatically offered to a waitlisted student instead of going unused — eliminating wasted professor slots.
- Layered architecture split across dao/, service/, and view/ packages, with SQL bootstrap scripts (schema.sql, sample-data.sql) for local setup.

AI Lounge Bot Shield — Multi-Signal Bot Detection

2024

Solo build, developed as part of TUKL-DL Lab internship | Python, FastAPI

- Built to protect the AI Lounge course-pricing website: competitor scraper bots would read live prices and feed them into their own dynamic pricing, shifting the market to the lab's disadvantage.
- Scores every request across four signals: rate limiting (req/min per IP), User-Agent heuristics (curl/wget/scrapy signatures), header-anomaly checks (missing Accept/Accept-Language/UA), and a JS-challenge cookie (HMAC-signed, IP+UA-bound, issued only after a real browser runs JS on /challenge).
- Requests scoring ≥ 3 are classified as bots and served decoy prices; legitimate traffic sees real prices. Known search-engine bots (Googlebot, Bingbot) can be allowlisted.
- Documented a known detection gap via a regression script (scripts/bypass_bot.py): a scripted client with realistic headers but no valid cookie can still slip through, and outlined the fix.

NUST FAQ Chatbot — 2nd Place, ACM SEecs Local Chatbot Competition 2026

2026

Solo build | Organised by Dr. Sohail Iqbal & Faisal Shafiat | NUST ACM Student Chapter Python, Flask, FAISS, Sentence-Transformers, llama-cpp-python

- Fully offline admissions assistant answering NUST student queries (admissions, MBBS, fees, scholarships) with zero internet dependency, runs entirely on CPU within 8GB RAM.
- Spell correction via SymSpell seeded with a custom NUST dictionary (handles typos/slang like "mbbs", "nshs", "mdcat").
- Semantic retrieval via Sentence-Transformers (MiniLM) embeddings indexed with FAISS over 73 official NUST FAQs.
- Answers constrained to retrieved context using a quantized Qwen-1.5B model (GGUF, CPU-only via llama-cpp-python) — a strict RAG pipeline that prevents hallucination.
- Multi-question splitting (queries joined by "and"/commas answered separately then combined) and a Flask-served chat UI.
- Placed 2nd out of all participating teams; submitted with a YouTube demo, GitHub repository, and full documentation.

Dynamic Pricing Agent — Ongoing

TUKL-DL Lab internship follow-on, 2024–present

Research context: optimal pricing, demand forecasting, and market-signal analysis | Python | Prophet, XGBoost, GluonTS, Darts, ARIMA, sentence-transformers, Apache Kafka

- Integrated **demand forecasting, cross-platform product matching, real-time price streaming, and bot detection** into a dynamic pricing pipeline using Walmart/Amazon market data.
- Currently extending the system into a standalone application that combines **competitor price matching, demand prediction, and dynamic pricing adjustments**.

No public repo yet — actively in development.

Portfolio Website

2025

Next.js 14, TypeScript, TailwindCSS, Framer Motion

- Personal portfolio site, vibe-coded and still being refined — dark/light theming, responsive layout, and animated sections built with Framer Motion.

Navigating the Future: The Role of Collaborative Robots (Cobots) in Industry 5.0

2024

Research Review

- Conducted a literature review examining the role of collaborative robots (cobots) in Industry 5.0, with emphasis on machine learning, adaptability, and decision-making.
- Analysed applications of cobots in manufacturing, including workplace safety, flexible automation, and workforce collaboration, alongside implementation challenges.
- Reviewed ethical considerations and case studies related to the adoption of cobots in industrial environments.

Sudoku Game in C++

- Terminal-based Sudoku with game logic implemented in C++.

DailyWord — Wordle-Style Game | C++, Python, Tkinter

- Developed a Wordle-style game integrating a C++ backend for game logic with a Python/Tkinter graphical interface.

ACHIEVEMENTS

- 2nd Place — ACM SEecs Local Chatbot Competition, NUST FAQ Chatbot — April 2026
- Certificate of Participation — NUST Mathematics Olympiad (NMO-2024) — 4 December 2024
- Certificate of Participation — Code O'Clock Workshop, NUST ACM Student Chapter — 20 November 2024
- Certificate of Participation — LaTeX for Academic Purposes, SEecs — August 2024

- Certificate of Participation — Comic Kahani Workshop (storyboarding, illustration) — 11 October 2024
- Certificate of Appreciation — Hope Pakistan Internship — September 2023
- Certificate of Recognition — EducationUSA High Achievers' Ceremony — 16 November 2023

SKILLS

Technical

- **Programming Languages** — Python (advanced), Java, C++, MySQL
- **AI & Machine Learning** — Large Language Models (LLMs), Transformers, Speculative Decoding, Hierarchical Speculative Decoding, NLP, Prompt Engineering, AI Evaluation, Model Benchmarking
- **Deep Learning Frameworks** — PyTorch, Hugging Face Transformers, TensorFlow (basic)
- **Machine Learning** — Supervised & Unsupervised Learning, Time Series Forecasting, NLP, Computer Vision (image-based similarity), Embeddings, Model Evaluation (Accuracy, Precision, Recall, F1, MAE, RMSE, MAPE)
- **Python** — data preprocessing, feature engineering, ML workflows (Prophet, XGBoost, GluonTS, Darts, ARIMA, sentence-transformers)
- **Data Streaming** — Apache Kafka (brokers, topics, producers/consumers, partitions, Avro serialization)
- **Backend** — FastAPI, Flask, REST APIs
- **Databases** — MySQL, SQL, MongoDB (Database Design, Normalization, Stored Procedures, Triggers)
- **Tools & Platforms** — Git, GitHub, Linux, Jupyter Notebook, Google Colab, Kaggle
- **Productivity Tools** — MS Excel (Intermediate) | MS Word & PowerPoint (Advanced) | LaTeX (beginner)
- **Designing Tools** — AutoCAD (intermediate) | Canva

Soft Skills: Attention to detail | Teamwork | Meeting deadlines | Critical thinking | Technical communication

Creative: Crochet & Embroidery — advanced

LANGUAGES

- English | Urdu

REFERENCES

Altaf Shah *Director and A-Level Maths Teacher | a.shah@sns.edu.pk*

Shabbir Jungee *Director and A-Level Chemistry Teacher | s.jungee@sns.edu.pk | +92 300 5115446*

Dr. Sohail Iqbal *Professor @ SEecs | Director of HPC Lab | sohail.iqbal@seecs.edu.pk*